

Assignment

Write your project proposal

Deadline : see Brightspace

Handout for the *Natural Computing* lecture, Proposing a project

This sheet will help you and your team brainstorm for project ideas and refine them into a project proposal. Part A is not obligatory; if you already have a great project idea, feel free to use that, and go directly to part B to refine it into a concrete proposal!

Objectives of This Exercise

1. Apply brainstorming techniques to obtain several ideas for potential projects (the more, the better!)
2. Select the most promising idea with your team and refine it to fit the scope of the project
3. Refine your idea into a concrete project proposal.

Part A: (optional) brainstorming exercises

Exercise A.1 Pick a focus topic (5 min)

Look back at the topics from the lectures and assignments and consider which one(s) you would like to explore in more depth. Each group member picks one. This will be your focus for the upcoming brainstorming session.

Exercise A.2 Brainstorming: part I - individually (20 min)

The best way to obtain good ideas is to first obtain many ideas (good ones and bad ones). Focus on quantity over quality – try not to judge or evaluate your ideas yet and Simply write down anything that comes to mind. You can use the following strategies, as well as the sources on Brightspace for inspiration:

1. **Extend/modify/follow-up** Look back to the lecture(s) and/or assignments on your focus topic. Can you think of interesting extensions/modifications to explore? Do any follow-up questions come to mind?
2. **New combinations** Can you make new combinations? For example, by combining your method with another method from the course (CPMs + evolution)?
3. **Comparisons** Could you compare your method to another method to understand in which condition(s) they work and/or how they differ? (E.g. spatial vs non-spatial SIR model, collective motion in CPM vs Boids, ...)
4. **Analogies** Take a problem you saw in the lectures/assignments. Can you think of analogous (real-world) problems or phenomena that you could apply the same method/model to? (E.g., spreading of infections → spreading of forest fires).
5. **Literature** You can also start with a paper from the scientific literature to see if any follow-up ideas come to mind (e.g. from the GECCO conference, the journal PLOS computational biology, or from the sources of inspiration listed on Brightspace).

Exercise A.3 Brainstorming: part II - individually (10 min)

Exchange your lists of ideas from exercise A.2 with another team member, and add related ideas/revised ideas/improvements.

Exercise A.4 Brainstorming: part III - with group (15 min)

Now that you all have a list of ideas, it is time to be more critical:

- Which ideas do/do not seem interesting to you?
- Which ideas are feasible (or could be made feasible)? Think of total scope, amount of work to implement, computational resources required, etc.
- Can you split the project into concrete tasks that align with the skills/background/interests of your team members, so that everyone can contribute?
- What are the pros and cons of your favorite ideas? Are there ideas that could be combined?

Use these questions to prune your list and select one or two ideas to continue with.

Part B: Refine your idea into a concrete proposal

Exercise B.1 Refine your idea and write a problem statement

Take your idea and discuss together how you can refine it into a concrete and specific problem statement:

1. Has this (or something similar) already been done (do some online searching)? If not, is it feasible to start this from scratch? If yes, can you extend or follow up on existing work to make a novel contribution?
2. In your problem statement, describe the background of this problem and what makes it interesting from a scientific perspective (look at related papers to see how they do this).
3. Write down specific goals and research questions to answer. (They should be broad enough that you can come up with a few experiments to answer them, but of course not so broad that you cannot complete the work during the course. Aim roughly for the scope of the previous assignments combined – reach out to the teachers if you are in doubt). Clearly define your problem/goals/questions in your problem statement.
4. Describe (roughly) how you will approach this problem and why this approach is suited to your task or question (again, you can look at examples from literature on how to phrase this).

Exercise B.2 Summarize related work

1. Find at least 3 "related works" from literature that are relevant to your project idea. For example, if you plan to apply method A to solve problem/question B:
 - Are there other works applying method A to problems similar to, but not quite the same as, problem B?
 - Did other works solve problem B with methods similar to, but not quite the same as, method A?
 - Are there other works that learned something about key features of method A or problem B that might be of interest to your work?
 - ... etc.
2. Your related works section should consist of one paragraph per (group of) related work, in which you:
 - Briefly summarize what they did (at least the part that is relevant to your project - the goal is not to summarize the paper but the specific "lessons learned" that you will build on). For example: *"Papers A, B and C applied D to E and found F."*
 - Make explicit how this relates to your project. What did they do that you will also do? What did they not do that you will now add? Why is that addition interesting/important? For example: *"Papers A, B and C applied D to E and found F. However, they did not test if this also does G, which is an important question because H. In this work, we also apply D to E, but now also look at G."*

Exercise B.3 Map out your approach and required resources

1. Outline the approach/algorithm(s) that you will use to tackle your problem, and justify this chosen approach. What properties of the problem and algorithm make this a good approach for the problem?
2. What resources do you need for your project, and how will you obtain them? For example:
 - Can you build on existing implementations or do you need to start from scratch?
 - Do you need any data, and where will you obtain that?
 - Do you need anything else? (For example: if you want to study infection spreading on a graph, how will you build your graph structure? Or will you obtain them from somewhere? Etc.)

Exercise B.4 Plan your milestones and supervision moments

1. What is your timeline? Think of the time you have available, and make a rough timeline for:
 - Implementing prototypes
 - Designing and executing experiments (this should be the focus!)
 - Analyzing outputs and thinking of ways to communicate them (you should reserve some time for this as well).

Build in “checkpoint” moments where you evaluate if you are on track and if any adjustments are necessary.

2. Assess feasibility again and improve if necessary. For example:
 - if you want to perform simulations that take a week to run, you will probably not manage to perform many experiments. Can you switch to a simpler simulation/algorithm/problem that takes less time to run?
 - Is there existing software you can build on? If you have to implement everything from scratch, be careful that you have enough time left for the experimentation phase. Simplify, if necessary.

Again, ask the teacher(s) if you are in doubt.

3. Have a look at the “checkpoint” assignments in Brightspace, and set (preliminary) goals for what you would like to discuss in each.

Product

Write a 2-page project proposal, in pdf, with the following elements:

- A (tentative) title for your project
- A problem statement with the background, relevance, and a concrete description of the problem you are solving (use exercise B.1 as input)
- A section “related work” describing the relation of your project to other work in this area (use exercise B.2 as input)
- A section “Approach” describing which methods you will use, why they are appropriate, which resources you need, and where/how you will obtain those (use exercise B.3)
- A section “Planning” where you describe which milestones you want to finish when (use exercise B.4).

Note that many of these elements will likely end up in your final report as well, so it is worth investing some time in this! Your final proposal should be concrete enough that it allows you to get started on the project.

Hand in your proposal on Brightspace per the deadline listed there.